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International Journal of Applied Linguistics

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ORIGINAL ARTICLE

Efficacy of Artificial Intelligence (AI) Voice Cloning in Phonetic Self-Imitation for L2 Pronunciation Training

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Received: 16 February 2025 | **Revised:** 7 September 2025 | **Accepted:** 13 September 2025

Funding: The authors received no specific funding for this work.

Keywords: AI | comprehensibility | L2 English | L2 fluency | phonetic imitation | phonetic self-imitation | pronunciation teaching | voice cloning | autoimitacja fonetyczna | imitacja fonetyczna | płynność w wymowie w L2 | Sztuczna Inteligencja (AI) | klonowanie głosu

Abstract

The idea of using a well-matched and personalized voice (a so-called golden speaker) in L2 acquisition enables learners to expand their pronunciation repertoire. Phonetic self-imitation, an accent-conversion method in which acoustic characteristics in native utterances are extracted and transferred into the learner's speech input, so that an L2 learner mirrors the one's voice synthesized with that of a native speaker, was first proposed over thirty years ago. Since then, a handful of studies have shown that self-imitation has been shown to be effective for L2 pronunciation improvement. However, this method requires continuous enhancements with new technological capabilities related to the development of neural networks and artificial intelligence (AI). This study examines the effectiveness of selected AI tools (Revoicer and Speechify) using voice cloning in phonetic self-imitation practice, aiming to investigate whether there is a correlation between this method and the level of L2 fluency and comprehensibility. In an eight-week pronunciation practice, 21 Polish learners of English performed self-imitation tasks three times a week (for a total of 45 min per week), involving imitation of AI-modified utterances using AI tools. Progress was assessed through pre-, post-, and delayed post-tests, evaluated using a seven-point Likert scale by native English speakers and well-experienced teachers of English. Results indicate a significant improvement in L2 fluency and comprehensibility among participants using AI-assisted phonetic self-imitation. The findings highlight the potential of integrating AI-driven phonetic self-imitation practice into L2 learning, offering new opportunities for L2 learners to improve their pronunciation skills and be able to work at their own time and pace.

Abstrakt

Koncepcja wykorzystania spersonalizowanego głosu (tzw. "złotego mówcy") w nauce języka obcego (L2) umożliwia uczącym się poszerzenie repertuaru wymowy. Autoimitacja fonetyczna to metoda polegająca na przenoszeniu cech akustycznych z wypowiedzi rodzimych użytkowników języka na mowę ucznia, tak aby mógł on odtworzyć własny głos zsynchronizowany z głosem native speakera. Pomysł ten pojawił się ponad trzydzieści lat temu, a liczne badania potwierdziły skuteczność tej metody w doskonaleniu wymowy L2. Metoda ta wymaga jednak ciągłego rozwoju i dostosowywania do nowych technologii opartych na sieciach neuronowych czy sztucznej inteligencji (AI).

W niniejszym badaniu oceniono efektywność wybranych narzędzi AI (Revoicer i Speechify), które wykorzystują klonowanie głosu w autoimitacji fonetycznej. Celem było zbadanie czy stosowanie tej metody wiąże się ze wzrostem stopnia płynności i zrozumiałości wypowiedzi w języku angielskim jako L2. W trakcie ośmiotygodniowego treningu wymowy 21 polskich studentów

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uczących się języka angielskiego jako L2 wykonywało zadania autoimitacji trzy razy w tygodniu (łącznie 45 minut tygodniowo). Zadania polegały na naśladowaniu zdań zmodyfikowanych za pomocą narzędzi AI. Postępy oceniano na podstawie testów przed treningiem, tuż po jego zakończeniu i kilka tygodni po treningu, a ocen dokonywali rodowici użytkownicy języka angielskiego i doświadczeni nauczyciele, korzystając z 7-stopniowej skali Likerta.
Wyniki wskazują na istotną poprawę płynności i zrozumiałości w języku L2 u uczestników korzystających z metody autoimitacji fonetycznej wspomaganej przez AI. Otrzymane dane potwierdzają potencjał tej metody, oferując nowe możliwości doskonalenia wymowy we własnym tempie i w dogodnym czasie.

1 | Introduction

Pronunciation proficiency is one of the most important elements of effective communication in a second language (L2), as it directly affects intelligibility, listener comprehension, and communicative success (Derwing and Munro 1997; Levis 2005; Derwing and Munro 2009; Isaacs 2014). Nevertheless, achieving native-like pronunciation skills proves challenging for most L2 learners, owing to phonetic and prosodic discrepancies between their first language (L1) and L2 (Munro and Derwing 1995; Levis 2005). In the 1980s, phonetic self-imitation emerged as an innovative approach that enabled learners to imitate their voices in a way that incorporated features of native language production (Repp and Williams 1987; Nagano and Ozawa 1990). This approach leverages the familiarity of the learner's voice while fostering the internalization of L2 phonetic patterns (Adank et al. 2009). One of the most important aspects of this method is the use of the "golden speaker," a synthesized voice that adds native pronunciation features to the students' vocal traits (Ding et al. 2019). Many researchers recognize this method as effective in phonetic learning. For example, Nagano and Ozawa (1990) claim that self-imitation based on the acoustic adjustment of learners' speech input reflects native-like pronunciation features while retaining their unique vocal identity. This personalized feedback is more effective than direct imitation of native speakers, as it increases familiarity and prompts more natural and sustainable learning (Bissiri et al. 2008; Kusz 2022).

Recent advancements in AI have transformed computer-assisted pronunciation training (CAPT), with AI-based voice cloning enabling learners to synthesize their voices and replicate native-like pronunciation features. Neural network models like RNNs and LSTMs enhance speech accuracy in accent, rhythm, and intonation (Zhang et al. 2019). Building on these innovations, AI voice cloning addresses two key gaps: limited research on AI in self-imitation and the need for scalable, personalized L2 learning tools (Chen et al. 2020; Ding et al. 2019). By examining tools like Revoicer and Speechify, this study expands CAPT research and highlights AI's potential to revolutionize pronunciation training. Given the increasing availability of AI tools with varying costs, understanding whether such differences influence learning outcomes is important for teachers and institutions considering scalable solutions. Therefore, comparing tools like Revoicer (paid) and Speechify (free) responds to a growing need for evidence-based selection in educational settings.

Building upon these objectives, the study is structured to systematically determine the efficacy of AI-assisted phonetic self-imitation in fostering L2 English pronunciation fluency and comprehensibility, assess learners' perceptions of AI voice cloning tools in pronunciation training, and investigate the correlation between learners' engagement with self-imitation and their progress in pronunciation skills. As AI voice-cloning technology becomes increasingly available, the choice of platform, including differences in cost, technical capacity, and output quality, may affect training outcomes. Investigating tool variation helps determine whether improvements are attributable to the phonetic self-imitation method itself or specific tool features.

2 | Literature Review

2.1 | Technological Foundations of AI Voice Cloning in Pronunciation Training

The main objective of voice cloning is to create synthetic utterances closely resembling those of a specific speaker, referred to as the "cloned speaker" (González Docasal and Álvarez 2023). The rising popularity of voice cloning can be attributed to recent technological advances in deep learning, especially in sequence-to-sequence acoustic modelling and generative neural vocoders, which have improved the flexibility and realism of cloned voices (Zhang et al. 2019).

As explained by Lenglet et al. (2025), voice cloning systems follow a two-stage process. In the first stage, an acoustic model, such as Tacotron 2 or FastSpeech, converts text into acoustic representations such as mel-spectrograms. In the second stage, a neural vocoder (e.g., WaveNet, WaveGlow, or HiFi-GAN) synthesizes natural-sounding speech waveforms from these acoustic characteristics (Oord et al. 2016; Kim et al. 2021). The models are conditioned on compact numerical representations of speaker identity—embeddings—which are sourced from two to five minutes of audio (Jia et al. 2018).

Numerous studies have investigated the efficacy of AI voice cloning in native-like manipulation. Wang et al. (2017) pointed out that cloned voices created with the use of Tacotron 2 not only approximated target prosodic patterns better than the learner's original recordings but also improved rhythm and pitch. Further work by Yamagishi et al. (2019) investigating the controllability of cloned output confirmed that synthesized speech can be dynamically adjusted for syllable duration, segmental precision, and pitch alignment. As a result, researchers can model highly specific phonetic features. Additionally, Hsu et al. (2025) showed

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that Mandarin-speaking English learners practicing with cloned feedback, even after only short-term exposure, had better gains in pitch range control and stress-timing accuracy than the control group.

The golden speaker builder (GSB), introduced by Ding et al. (2019), leverages this technological advancement for L2 pronunciation training by integrating a learner's voice with native-like prosodic characteristics. It aligns the learner's original speech with native utterances using dynamic time warping and pitch-shifting algorithms. As a result, not only does the modified output preserve voice identity, but it also adopts native rhythm and intonation. Such a synthesized voice is used by the learner as a pronunciation model.

Although tools based on voice cloning remain largely experimental, a growing body of research indicates the potential of AI to transform CAPT through personalized, adaptive, and scalable learning contexts (e.g., Felps et al. 2009; Shufang 2021).

2.2 | Voice Cloning as a Tool for L2 Pronunciation Acquisition

According to Ding et al. (2019), when paired with phonetic self-imitation, voice cloning contributes to the improvement of native-like L2 pronunciation by offering personalized feedback in the form of the learner's voice adapted to both target-like prosodic and segmental features. AI-generated feedback, by retaining the learner's vocal identity, improves familiarity, self-monitoring, and motivation (Adank et al. 2009; Pellegrino and Vigliano 2015; Saito and Plonsky 2019). Such personalized modelling has also been shown to increase learners' engagement (Mroz 2020).

While this study investigates the potential of personalized AI-assisted feedback, we acknowledge that traditional classroom-based pronunciation instruction is also effective (e.g., Derwing and Munro 2005). The goal of AI-enhanced self-imitation is not to replace teacher-led instruction, but rather to offer a flexible and scalable complement that supports individual practice, especially outside the classroom.

In second language acquisition (SLA) terms, hearing one's AI-modified voice functions as implicit feedback. Learners perceive phonetic differences without direct instruction, fostering internalization (Ellis 2009; Goo and Mackey 2013), which is in line with input enhancement models and reformulation feedback that encourage processing at the learner's own pace (Loewen and Nabei 2007; Lyster and Saito 2010). Cognitive processes underpinning this mechanism involve sensorimotor learning and output-driven modification by which learners modify their production based on perceived discrepancies with a native-like version of themselves (Houde and Jordan 2002; Swain 2005). Earlier systems also experimented with learners' own voices in corrective feedback, indicating that such practice can support intonation and pitch accuracy (Hirose et al. 2003; Peabody and Seneff 2006).

Research on memory shows that familiar voices are easier to process and remember. Goldinger and Azuma (2004) demonstrated that L2 learners process familiar voices more efficiently. Eger and Reinisch (2019) found that German learners of English recognized words more easily in their own recorded voices than in unfamiliar ones. The "own-voice effect" has also been documented in bilingual contexts. Cheung and Babel (2022) found that early bilinguals recognized words faster and more accurately when they were spoken in their own voices. Additionally, in a study on English intonation training, Li and Lian (2022) showed that imitating a model voice reduces phonological memory load and helps L2 learners focus more on articulation and prosody. This indicates that AI-based self-imitation tools may be effective not only because of technology but also because familiar voices offer cognitive advantages.

From a pedagogical perspective, several studies have examined the efficacy of AI voice cloning in native-like pronunciation acquisition in L2 learners. For example, in Ding and colleagues' study, L2 learners were assigned to two groups: a GSB-based self-imitation group and a native-speaker imitation group. After three training sessions, the GSB group outperformed the native-speaker imitation group in terms of comprehensibility, fluency, and prosodic control, pointing to personalized voice feedback offering stronger phonological alignment than imitation of external models. These improvements were especially evident in rhythm, stress placement, and intonation patterns, all of which are crucial for intelligibility and naturalness (Hardison 2004; Trofimovich and Isaacs 2012). Personalised pronunciation training with phonetic self-imitation has also been found effective in improving prosody in L2 learners (Pellegrino and Vigliano 2015). These benefits were confirmed by Saito and Plonsky (2019). According to the authors, when learners compare their unmodified voice to a target-like, AI-generated version of themselves, they receive perceptually salient, individualized input that enhances deeper processing of phonological input and more durable learning effects. This aligns with sensorimotor learning theory, which posits that speech development is guided by internal auditory models used to adjust output over time (Houde and Jordan 2002; Shiller et al. 2009).

Finally, research in auditory cognition also highlights the importance of voice familiarity. Holmes et al. (2018) showed that one's own voice, as the most familiar sound, is easier to process and thus helps retain pronunciation. This supports the use of AI voice cloning in pronunciation training.

2.3 | Research on L2 Pronunciation Outcomes: Comprehensibility and Fluency

Comprehensibility and fluency are widely acknowledged in L2 pronunciation research as instructionally sensitive and communicatively relevant (Derwing and Munro 2005; Trofimovich and Isaacs 2012). Comprehensibility is defined as the listener's ease of understanding (Munro and Derwing 1995), whereas fluency includes temporal features such as speech rate, pausing, and hesitation (Derwing et al. 2004).

Unlike traditional measures such as accentedness or segmental accuracy, which often prioritize native-likeness, comprehensibility, and fluency focus on successful communication (Levis

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2005). Studies show that segmental deviations have less impact on perceived comprehensibility than suprasegmental features such as stress placement and intonation (Trofimovich and Isaacs 2012). Similarly, Saito et al. (2016) found that prosodic control, not accent reduction, was a stronger predictor of both comprehensibility and perceived fluency in L2 learners, supporting the view that fluency and comprehensibility serve as ecologically valid indicators of learner progress, especially in contexts where intelligibility is prioritized over nativeness.

Moreover, comprehensibility and fluency are especially responsive to prosody-based training targeting rhythm, timing, and intonation, which aligns with the AI-assisted self-imitation objectives (Hardison 2004; Bissiri and Pfitzinger 2009). Since voice cloning systems like the GSB are designed to correct suprasegmental features while preserving speaker identity, comprehensibility and fluency are appropriate and theoretically grounded outcome measures.

In the present study, the read-aloud data were used to evaluate pronunciation progress across trained and untrained sentences. While we acknowledge the value of spontaneous speech in capturing real-time processing and less monitored production (Kormos 2006; Tavakoli and Skehan 2005), we focused on read-alouds for this initial analysis to ensure task consistency, comparability across participants, and alignment with the training input, which was also sentence-based (Derwing et al. 2004; Munro and Derwing 1995). Spontaneous speech often introduces greater variability in syntax, lexical choice, and topic complexity, which can reduce the reliability of listener judgments in small-sample studies (Kang et al. 2010; Isaacs and Thomson 2013). Nevertheless, spontaneous speech samples were collected and may be analyzed in future work to explore generalization and transfer effects beyond trained stimuli (Segalowitz 2016).

In summary, the selection of comprehensibility and fluency as outcome variables is based on both theoretical grounding and empirical precedent, reflecting a shift in pronunciation research toward intelligibility-based instruction (Levis 2020). These constructs not only capture changes in listener-perceived speech quality but also serve as practical proxies for communicative effectiveness, especially when investigating the impact of prosody-targeted, AI-enhanced self-imitation.

3 | Research Questions

The study was guided by the following research questions:

RQ1: To what extent does AI-assisted phonetic self-imitation practice improve L2 learners' comprehensibility and fluency over time?

RQ2: Does the choice of AI voice-cloning tool (Revoicer vs. Speechify) influence L2 learners' comprehensibility and fluency?

RQ3: To what extent are improvements in L2 comprehensibility and fluency maintained over time following AI-assisted phonetic self-imitation training?

4 | Methodology

4.1 | Participants

A total of 25 participants took part in this study. The experimental group consisted of 21 Polish learners of English (19 females, 2 males) who were students of Applied Linguistics at a university located in the southeastern part of Poland. Their age ranged from 19 to 23, and they all completed a 30-h course in English phonetics and phonology with positive final test results, which indicates they were familiar with the basic concepts of English phonetics and phonology, as well as English and Polish sound systems, including their similarities and differences. A 30-hour English phonetics and phonology course covered both segmental and suprasegmental aspects of English pronunciation, including stress patterns, rhythm, intonation, and connected speech processes. Students' proficiency in English varied from B2 to C1, as determined by their final standardized practical English exams. The participants had no hearing impairments or speech disorders. L2 learners were recruited through convenience sampling, i.e., we gathered data from all students who were willing to participate in the study.

In addition, a control group of four participants (3 females, 1 male) was recruited from the same student population. The control group did not engage in phonetic self-imitation training but participated in the pre- and immediate post-test. These students were included to compare the results and to examine whether any improvements could occur without the training, for example, due to regular English classes or exposure to English over time.

The number of participants in the control group was limited, as all students participated in the study on a voluntary and unpaid basis. While participants in the control group completed the pre- and immediate post-test, they did not submit the delayed post-test recordings, despite several reminders and requests. As a result, delayed post-test data were only available for the experimental group.

4.2 | Tools and Materials

For this study, two AI tools were employed, namely Revoicer and Speechify. Both are AI-powered text-to-speech tools designed to allow users to create high-quality cloned voices. While there are currently many paid AI tools available that offer users voice cloning features, we selected Revoicer (paid) and Speechify (limited free version) as representative examples for this study due to their accessibility and pricing. The choice to implement these AI tools, despite the small number of participants in each group (Speechify $n = 10$, Revoicer $n = 11$), was intended to explore their potential effects on L2 pronunciation. Their cloning capabilities were used to create a "golden speaker" model, which was a synthesised native English voice that was matched to each participant's vocal characteristics. Only the project coordinators had access to the premium versions of these tools, and they manipulated students' recordings to provide them with their golden speaker/model voices. Both Revoicer and Speechify used AI technology based on neural network models for voice synthesis and cloning. The voice cloning process was

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based on the students' recordings from the pre-test. A three- to four-minute sample from each participant was input into both tools, which resulted in new voice samples. The voice samples were based on script materials previously prepared by the project coordinators (see Appendix 2). The sentences used in the tests (particularly the tongue twisters included in task one) were adapted from course books on English pronunciation (e.g., Mańkowska et al. 2009). The same sentences were used in the pre- and post-tests. Although the order of tasks and sentences remained the same, the eight-week interval between the pre- and post-test was expected to minimize possible practice effects, particularly because only half of the sentences were used during training. The others appeared only in the tests. This helped evaluate both trained and untrained material. No distractors were included, as the tasks were intentionally designed to focus on target sounds, which helped evaluate the effectiveness of AI-driven phonetic self-imitation on L2 comprehensibility and fluency.

4.3 | Experimental Design and Procedure

A total of 25 participants were asked to take a pre-test, which included 30 tongue-twister sentences, a four-paragraph text (165 words), and three open-ended questions. The tasks were designed on the basis of pronunciation challenges for Polish speakers learning English as a second language (Czaja 2024, Nowacka 2019; Porzuczek 2015; Sobkowiak 1996). Students were asked to submit a 5–7 min recording. These pre-test recordings were then used by the project coordinators to create model voice samples in either Revoicer or Speechify. The participants were divided into two groups, based on the AI tool implemented in this study, namely Speechify ($n = 10$) and Revoicer ($n = 11$). The participants were randomly assigned to practice with a single tool and they were not allowed to switch during the pronunciation training sessions.

The main part of the study was an eight-week training session, during which the participants were asked to engage in self-imitation practice three times a week, with each session lasting 15 min, totaling 45 min per week, and 6 h of pronunciation practice overall. Every pronunciation training session followed this procedure: (1) participants listened to their AI-generated model; (2) they focused on one sentence or phrase at a time; and (3) they paused the recording and repeated the chosen fragment as many times as necessary, concentrating on specific sounds, intonation, rhythm, or speed of a sentence; The key rule was to avoid mindless mimicry. Instead, they were instructed to practice consciously and with intention. Students were encouraged to carefully listen to the AI-generated model voice and, while imitating, focus on accuracy in segmental features such as consonants and vowels, particularly during the tongue twister task. In the second task, which required sentence imitation, they were instructed to pay attention to suprasegmental features such as rhythm, intonation, and speech rate. After each practice session, students were required to upload their recordings, which allowed the project coordinators to monitor their training and confirm their engagement; (4) once they felt confident with the current fragment, they proceeded to another one. No explicit corrective feedback was given, as the main goal of this experiment was to examine the effectiveness of phonetic self-imitation training rather than direct feedback.

Participants' performance was measured across three different time points: pre-test, immediate post-test (one week after the training session), and delayed post-test (four weeks after the training session ended). The students recorded their speech at home using either a smartphone or a computer. All recordings, including those from the pre- and post-tests as well as the practice sessions, were submitted via the Microsoft Teams platform. Although the recording environment was not controlled, the students were instructed to record in a quiet space to minimize background noise. Project coordinators reviewed each recording to ensure it was clear and could be included in the dataset. Before the training session began, the students were carefully instructed on how to do the self-imitation practice. It was a listen-and-read task that students completed at their own pace. They could imitate the voice sample they received as many times as they wanted, however, each training session took them exactly 15 min of focused pronunciation practice. The cloned voice sample they received each week was a 2–3 min recording that consisted of three tongue twisters and a short paragraph.

Since participants did the self-imitation task at their own pace, they were required to record each session and submit it to the project coordinators as proof of task completion. Students were not permitted to complete all training sessions in one day. They were explicitly instructed to practice three times a week. The first training session could be done between monday and tuesday, the second between wednesday and thursday, and the last training session between friday and sunday. This procedure ensured that participants maintained consistency in their pronunciation practice, which was one of the main aims of the experiment. Project coordinators reviewed selected fragments from all participants' recordings to ensure they followed the procedure. If the students had any issues, the project coordinators addressed them as quickly as possible.

4.4 | Assessment and Evaluation

22 raters participated in the assessment process. They were either native English speakers who were also teachers of English as a second language ($n = 7$) or well-experienced Polish teachers of English with more than 10 years of experience ($n = 15$). Their task was to assess the recordings using a seven-point Likert scale in two categories: comprehensibility and fluency. Comprehensibility and fluency scores were calculated as the average ratings given by all raters for each audio sample. To ensure rating consistency and reliability, inter-rater agreement was later evaluated using the intraclass correlation coefficient (ICC), which showed excellent consistency across raters (ICC = 0.99 for comprehensibility and ICC = 0.99 for fluency; see section 5.1). Then, mean scores were calculated for each participant at the pre-test, immediate post-test, and delayed post-test stages. By comprehensibility, we mean how much effort a listener has to exert to understand a speaker's message. Fluency, on the other hand, 'pertains to the ability to speak with minimal hesitation, unnatural pauses, or disruptions. Both categories were rated by all the raters. After segmenting the recordings obtained from the pre-, post-, and delayed post-tests

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into short audio files lasting several seconds, a total of 2813 audio samples were generated. Since the total duration of the recordings exceeded 14 h, it was not feasible for every rater to assess every recording. Therefore, we limited the number of recordings to 320, ensuring that all raters evaluated the comprehensibility and fluency of each participant in the study. The audio samples were grouped into eight surveys, each containing 40 audio samples. Each survey took approximately 20 min to complete, and raters had two weeks to accomplish all tasks.

All raters received clear written instructions before beginning the rating task. These were available in the introductory part of the survey. The instructions provided definitions of comprehensibility and fluency based on previous research (Derwing and Munro 1997; Trofimovich and Isaacs 2012). The instructions also included a detailed explanation of the seven-point Likert scale. Each point on the scale was described in relation to comprehensibility and fluency. For comprehensibility, a rating of 1 indicated that the speaker was difficult to understand, whereas 7 meant the speaker's speech was very easy to comprehend. For fluency, 1 represented very disfluent speech, with many pauses, hesitations, and unnatural breaks, whereas 7 indicated a high level of fluency, with smooth and natural speech. Although we did not collect any trial ratings, all raters were required to read the instructions carefully before beginning the rating process. No collective training or calibration session was conducted prior to rating; however, all raters were experienced university-level English instructors familiar with pronunciation assessment protocols. If the raters encountered any issues, they were encouraged to contact the project coordinators.

First, they were asked to familiarize themselves with the concepts of comprehensibility and fluency. Then, they were instructed on what the numbers on a 7-point Likert scale represent. This was to ensure consistency in their evaluation. A seven-point Likert scale was used for comprehensibility ratings, following previous research that demonstrated its clarity and usefulness in L2 pronunciation assessment (e.g., Derwing and Munro 1997; Isaacs and Trofimovich 2012).

Comprehensibility and fluency were rated by the same group of raters ($n = 22$). The average length of each recording was 5–7 min. All tests included 14 tongue twisters, which occurred during the eight-week training session (trained sentences/phrases). There were also 16 untrained sentences that the students were asked to read and record, which did not occur in the training sessions. Similarly, the second task which was reading a paragraph consisted of seven trained sentences (121 words) and two longer, untrained sentences (44 words) (see Appendix 1). The last task involved extemporaneous speech, which lasted approximately 3 min on average. However, this part of the test was not assessed by the raters, as scripted speech provided a more controlled and comparable measure of L2 pronunciation improvement. Moreover, given the already large number of scripted sentences/phrases included in the assessment, adding voice samples from extemporaneous speech would have significantly increased the raters' cognitive load and assessment time. This, in turn, could have compromised rating reliability.

In the tongue-twister task, all sentences were first extracted and then categorized as either trained or untrained. From the paragraph section, we excluded a 30 s sample from the trained part and a 20 s sample from the untrained part. Our main goal in the evaluation process was to ensure that each rater assessed every participant. Thus, we designed eight surveys, each including at least one recording from every participant. One rating session consisted of 40 voice samples, typically taking about 20 min. Each rater received eight links, with different voice samples in each link. Raters were asked to assess students' recordings on a 1-7 Likert scale, where 1 represented the lowest rating, and 7 the highest, for each dependent variable. Raters had two weeks to complete the rating of all the links received. Project coordinators monitored the raters' progress and provided immediate assistance if any issues occurred.

4.5 | Data Analysis

In data analysis, we focused on our two main research objectives. First, we aimed to assess the effectiveness of an AI-assisted self-imitation technique in improving L2 English comprehensibility and fluency. Second, we sought to compare the performance of two AI tools, namely Revoicer and Speechify, used in the study.

Inter-rater reliability was assessed using the ICC, which is a correlation-based metric. In a reliable rating process, if one rater evaluates a recording X higher than recording Y, other raters scoring the same recordings are also expected to assign a higher value to recording X. Our dataset yielded a high ICC = 0.99), which indicated excellent agreement among raters. Given this high reliability, the analyses were performed based on the collected ratings. The dependent variables were comprehensibility and fluency scores. Independent variables were Time (pre-test, immediate post-test, delayed post-test), Training (trained vs. untrained phrases/structures), Tool (Revoicer vs. Speechify), and Group (experimental and control).

Before conducting statistical analyses, the normality assumption for each dependent variable was checked using the Shapiro-Wilk test. It confirmed that comprehensibility and fluency ratings followed a normal distribution. To address the research questions and compare AI tools and their effectiveness in L2 pronunciation improvement we conducted:

1) A repeated-measures ANOVA to analyze within-subject changes over time (pre-test, post-test, delayed post-test) in the experimental group.
2) Independent-samples t-tests to examine group differences between the experimental and control groups.
3) Independent-samples t-tests to compare the effectiveness of AI tools (Revoicer vs. Speechify).

To identify pairwise differences between pre- post- and delayed post-tests, we also performed post-hoc analyses using Tukey's HSD test. Since the control group did not record delayed post-tests, we decided to exclude it from analyses requiring a time effect.

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TABLE 1 | Tukey HSD post-hoc comparisons for comprehensibility scores.

Comparison	Mean difference (*M*diff)	95% CI (lwr, upr)	*p*-value
Post-test vs pre-test	−0.62	(−1.08, −0.17)	0.005
Delayed post-test vs. pre-test	−0.35	(−0.81, 0.10)	0.163
Delayed post-test vs post-test	0.27	(−0.19, 0.73)	0.343

Note: Mdiff = mean difference; CI = confidence interval; lwr = lower bound; upr = upper bound; p = probability value. Results are from Tukey's HSD post-hoc test.

TABLE 2 | Paired-samples *t*-tests for comprehensibility (trained vs. untrained sentences/phrases).

Time Point	*t*-value	df	*p*-value	95% CI	Mean Difference (*M*diff)
Pre-test	4.04	24	< 0.001	(0.07, 0.19)	0.12
Post-test	2.91	24	< 0.01	(0.07, 0.35)	0.20
Delayed post-test	2.48	15	< 0.05	(0.04, 0.56)	0.30

Note: t = t-value; df = degrees of freedom; CI = confidence interval; Mdiff = mean difference.

5 | Results

5.1 | RQ1: Effect of AI-Assisted Phonetic Self-Imitation on Comprehensibility and Fluency Over Time

To answer RQ1, we examined whether AI-assisted phonetic self-imitation practice improved learners' comprehensibility and fluency after training.

A total of 7040 comprehensibility ratings were obtained from 22 raters. Each recording was assessed by each rater. Inter-rater reliability was measured using correlations between ratings (ICC = 0.99). The high *ICC* value indicates excellent agreement among raters, which confirms the reliability of the comprehensibility ratings and justifies their statistical analysis. The Shapiro-Wilk test results indicated that comprehensibility ratings were normally distributed ($W(24) = 0.97$, $p > 0.05$).

To explore whether there was an improvement in fluency, a total of 7,040 fluency ratings were obtained from 22 raters. Similarly to comprehensibility measures, each recording was assessed by each rater. The ICC (= 0.99) indicated excellent agreement among raters. Fluency ratings were normally distributed ($W(24) = 1.00$, $p < 0.001$).

5.1.1 | Comprehensibility (Time and Training effects)

To assess whether phonetic self-imitation supported by AI tools improved comprehensibility, a repeated-measures ANOVA was conducted. We compared ratings across three time points, namely pre-test, post-test, and delayed post-test. The results indicated a significant main effect of time, $F(2, 30) = 29.44$, $p < .001$. The results of Mauchly's test showed a violation of the sphericity assumption ($W = 0.40$, $p = 0.002$), thus a Greenhouse-Geisser correction was performed ($p[GG] < 0.001$).

Post-hoc analyses were conducted using Tukey's HSD test and paired-samples *t*-tests to further investigate differences between the results of pre- and post-tests. Table 1 presents the Tukey HSD results.

The results indicate that comprehensibility ratings significantly improved from pre-test to post-test ($p = 0.005$). However, there was no significant difference between the ratings of post- and delayed post-tests ($p = 0.343$), nor between the pre-test and delayed post-test ($p = 0.163$). Figure 1 presents comprehensibility ratings at pre-, post-, and delayed post-test stages. The results indicate improvement after AI-driven phonetic self-imitation training.

As seen above, AI-assisted phonetic self-imitation practice significantly improved L2 comprehensibility. The most prominent ratings were observed in the post-test scores. Although there was a decrease in delayed post-test results, L2 comprehensibility remained significantly higher than in the pre-test ratings, which suggests beneficial and long-lasting effects of self-imitation training.

To examine differences between trained and untrained sentences on L2 comprehensibility, paired-samples *t*-tests were conducted. We compared trained and untrained sentences/phrases at three time points: pre-, post-, and delayed post-tests. Table 2 presents the results of this part of the study.

As shown above, trained sentences/phrases received significantly higher comprehensibility ratings than untrained ones across pre-, post-, and delayed post-tests. Interestingly, trained sentences/phrases achieved better results even in the pre-test ($p < 0.001$). This might be due to the frequency and popularity of the items included in the trained sentence/phrase section. The most significant difference between trained and untrained sentences/phrases was observed in the post-test comprehensibility ratings ($p < 0.01$). This suggests that AI-assisted phonetic self-imitation practice is an effective method for improving L2 comprehensibility. Moreover, the effect persisted, as the results of the delayed post-test also indicated significantly higher comprehensibility ratings for trained sentences/phrases compared to untrained ones ($p < 0.05$). Figure 2 illustrates the distribution of comprehensibility scores for trained and untrained sentences/phrases across pre- post- and delayed post-tests.

Trained sentences/phrases consistently received higher ratings than untrained ones. The difference between trained and untrained sentences/phrases at the pre- and post-tests remained statistically significant. While the ratings slightly decreased in the delayed post-test, they remained higher than in the pre-test. Interestingly, Figure 2 also illustrates an improvement between the pre-, post-, and delayed post-test for untrained sentences/phrases. This confirms the effectiveness of AI-assisted phonetic self-imitation not only for trained structures but also for untrained ones.

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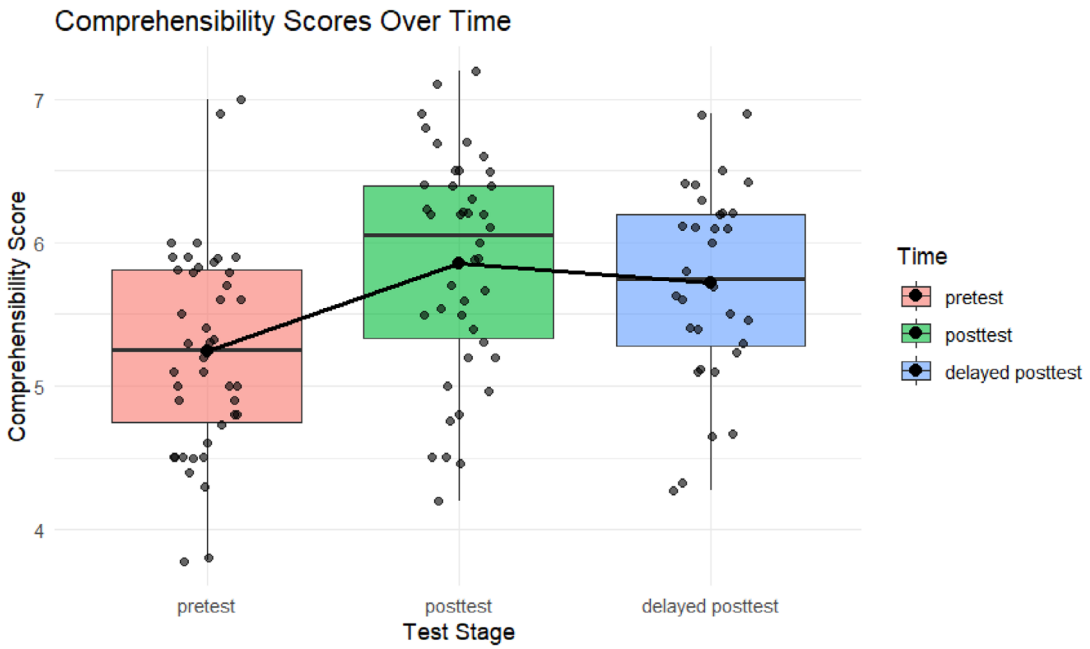

FIGURE 1 | Boxplot of comprehensibility ratings over time (experimental group). Each participant has two scores at each stage (trained and untrained sentences), shown together. [Color figure can be viewed at wileyonlinelibrary.com]

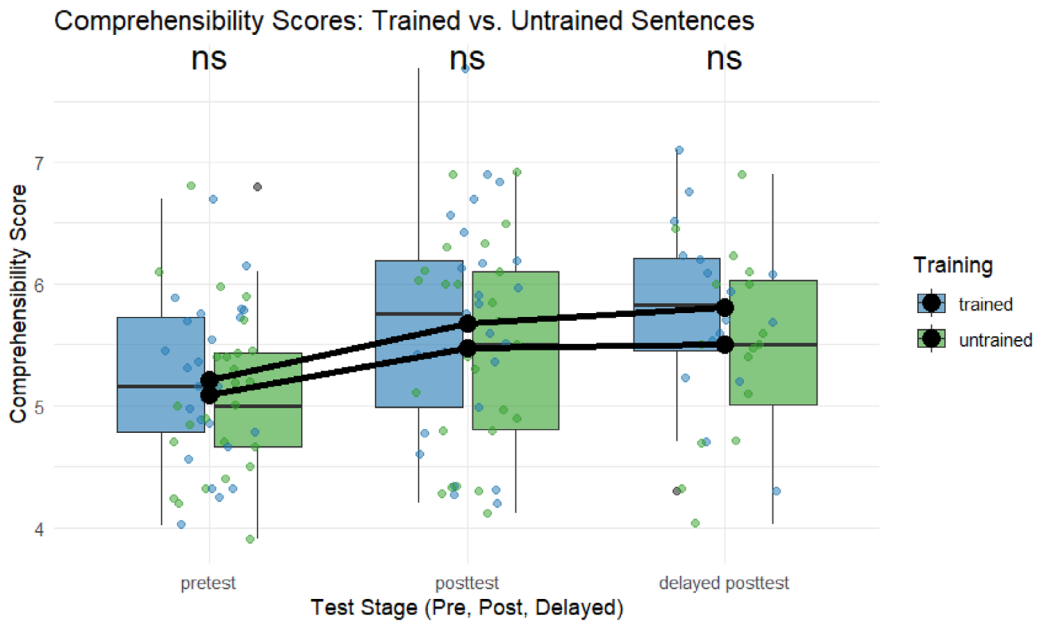

FIGURE 2 | Boxplot of comprehensibility ratings for trained vs. untrained sentences/phrases. [Color figure can be viewed at wileyonlinelibrary.com]

5.1.2 | Fluency (Time and Training effects)

The results of a repeated-measures ANOVA presented in Table 3 showed a significant effect of time on fluency ratings, F (2, 30) = 44.64, $p < 0.001$. This indicates that the level of fluency improved over time. The results of Mauchly's test revealed that the assumption of sphericity was violated ($W = 0.50$, $p = 0.008$); thus, the Greenhouse-Geisser correction was applied.

The results of post-hoc Tukey HSD comparisons confirmed that there was a significant improvement in L2 fluency between pre-test and post-test ratings (Mdiff = 0.71, $p < 0.001$), as well as between pre-test and delayed post-test (Mdiff = 0.44, $p = 0.043$). There was no significant difference between post-test and delayed post-test ratings (Mdiff = −0.26, $p = .317$), indicating that the participants of the study maintained their level of L2 fluency over time (see Table 4 and Figure 3).

The results of paired-samples t-tests revealed a significant improvement in L2 fluency for trained sentences/phrases compared to untrained ones in the post-test ratings ($p < 0.001$). The difference, although not as prominent, between trained and untrained sentences also occurred in the delayed post-test ($p < 0.001$) (see Table 5 and Figure 4).

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TABLE 3 | Results of repeated-measures ANOVA for fluency.

Effect	DFn	DFd	SSn	SSd	*F*	*p*	η^2 (ges)
(Intercept)	1	15	1538.90	22.90	1007.84	< 0.001	0.98
Time	2	30	4.11	1.38	44.64	< 0.001	0.15

Note: dfn = numerator degrees of freedom; dfd = denominator degrees of freedom; SS = sum of squares; η^2 (ges) = generalized eta squared.

TABLE 4 | Pairwise comparisons for fluency (Tukey's HSD).

Comparison	Mean Difference (*M*diff)	95% CI (Lower, upper)	*p*-value
Post-test—pre-test	0.71	(0.28, 1.14)	< 0.001
Delayed post-test—pre-test	0.44	(0.01, 0.88)	0.043
Delayed post-test—post-test	−0.26	(−0.70, 0.17)	0.317

Note: Mdiff = mean difference; CI = confidence interval; HSD = Honestly Significant Difference.

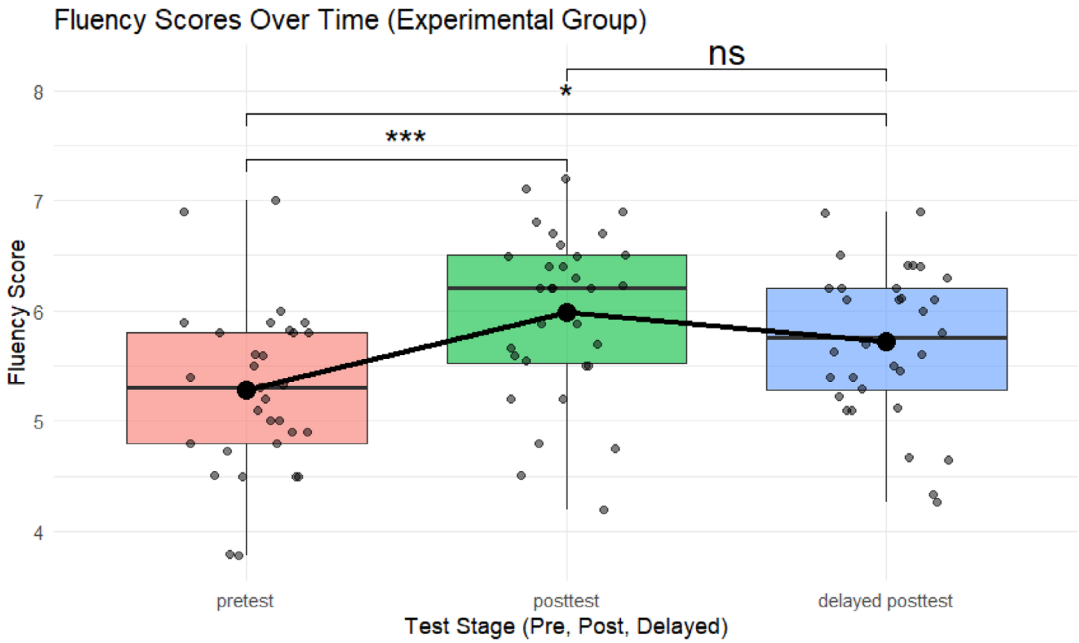

FIGURE 3 | Boxplot of fluency ratings over time (experimental group). Each participant has two scores at each stage (trained and untrained sentences), shown together. [Color figure can be viewed at wileyonlinelibrary.com]

5.1.3 | Group Comparisons (Experimental vs. Control)

We conducted independent-samples *t*-tests to examine the effect of group (experimental vs. control) on L2 comprehensibility. The results indicated statistically significant differences between experimental and control groups in pre-test and post-test. Delayed post-test was not taken into consideration in this part of the study, as the control group only recorded pre- and post-tests.

The results presented in Table 6 showed that the experimental group achieved better results even in the pre-test ($p = 0.015$); however, the difference was not prominent. This might be due to the small number of participants in the control group compared to the experimental group. In the post-test, the experimental group improved significantly ($p < 0.001$). This again confirms that AI-assisted phonetic self-imitation benefits L2 comprehensibility. As shown in Figure 5, the experimental group outperformed the control group, especially in the post-test comprehensibility ratings, suggesting the potential effectiveness of AI-assisted phonetic self-imitation practice.

The results of independent-samples *t*-tests presented in Table 7 indicate that there was a significant difference between the performance on L2 fluency in experimental and control groups in the post-test ratings ($p < 0.001$). Similarly to the L2 comprehensibility outcomes, the experimental group outperformed the control group significantly even in the pre-test ($p = 0.002$). However, this result was probably due to the discrepancy in the number of participants in the experimental and control groups. Figure 6 illustrates the findings, showing an evident and significant increase in L2 fluency ratings in the experimental group compared to the control group in the post-test ($p < 0.001$).

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TABLE 5 | Paired-samples t-test results for fluency (trained vs. untrained sentences/phrases).

Comparison	Mean difference (*M*diff)	95% CI	*t*-value	df	*p*-value
Post-test: trained vs. untrained	0.14	[0.10, 0.18]	7.10	24	< 0.001
Delayed post-test: trained vs. untrained	0.12	[0.06, 0.18]	4.27	15	< 0.001

Note: Mdiff = mean difference; CI = confidence interval; t = t-value; df = degrees of freedom.

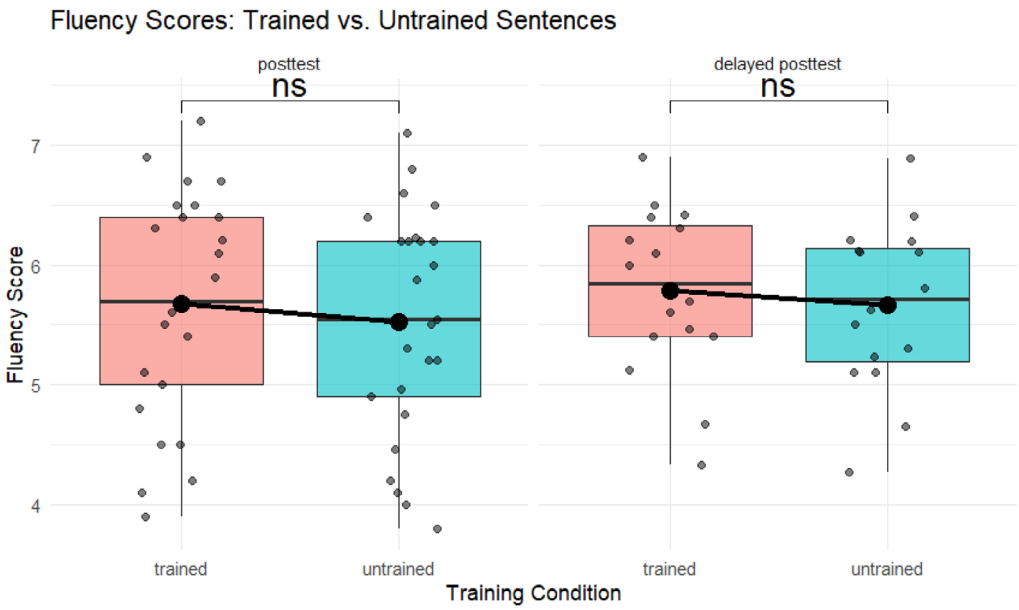

FIGURE 4 | Boxplot of fluency ratings for trained vs. untrained sentences/phrases. [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 6 | Mean comprehensibility scores for the experimental and control groups in the pre-test and post-test.

Test Stage	Mean (experimental)	Mean (control)	Mean difference (*M*diff)	95% CI	*t*-value	df	*p*-value
Pre-test	5.24	4.68	0.55	[0.13, 0.98]	2.80	12.83	0.015
Post-test	5.76	4.64	1.12	[0.68, 1.55]	5.39	17.82	<0.001

Note: Mdiff = mean difference; CI = confidence interval; t = t-value; df = degrees of freedom.

5.2 | RQ2: Effect of Tool Choice (Revoicer vs. Speechify)

5.2.1 | Effect of Tool on Comprehensibility

To address RQ2, we compared the effectiveness of Revoicer and Speechify. Independent-samples *t*-tests were conducted to examine which AI tool, Revoicer or Speechify, helped to achieve better results in L2 comprehensibility. The results indicated no significant differences between these tools at any stage.

Table 8 and Figure 7 illustrate that there was no significant difference between Revoicer and Speechify ($p = 0.629$) in the post-test and delayed post-test ($p = 0.948$), however, both AI tools significantly improved L2 comprehensibility.

The results of this part of the study indicate that AI-assisted phonetic self-imitation practice has a positive effect on L2 comprehensibility, regardless of the chosen AI tool.

5.2.2 | Effect of Tool on Fluency

The results of independent-samples t-tests at three different time points (pre-test, post-test, and delayed post-test) revealed that both tools led to improvements in L2 fluency, however, it is

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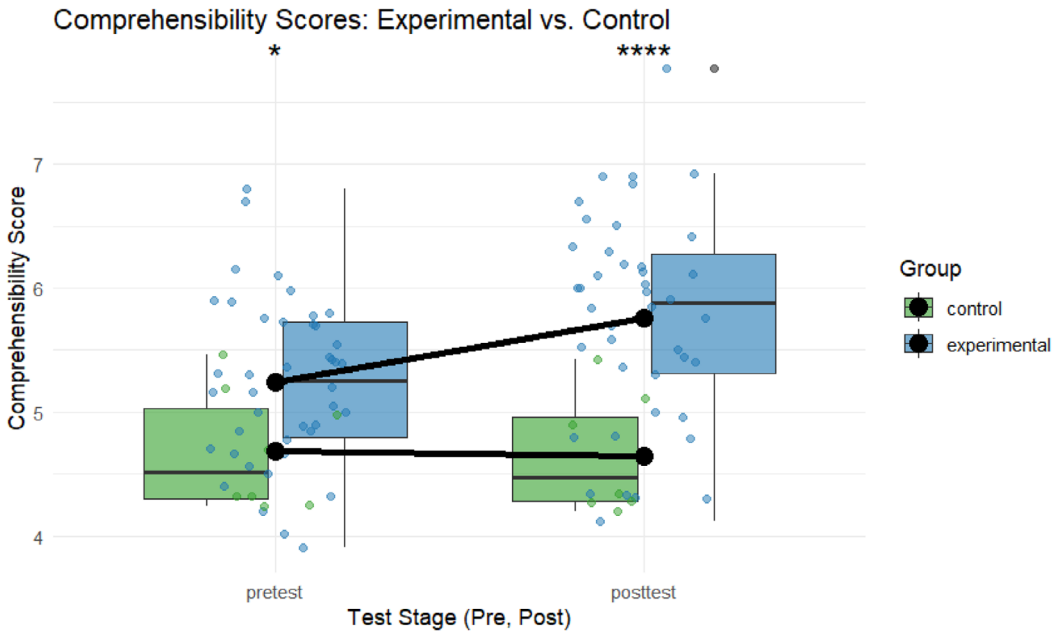

FIGURE 5 | Boxplot of comprehensibility scores in experimental vs. control groups. [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 7 | Mean fluency scores for the experimental and control groups in the pre-test and post-test. .

Comparison	Mean difference (*M*diff)	95% CI	*t*-value	df	*p*-value
Pre-test: experimental vs. control	0.81	[0.37, 1.25]	3.94	13.50	0.002
Post-test: experimental vs. Control	1.65	[1.15, 2.03]	7.73	15.21	< 0.001

Note: Mdiff = mean difference; CI = confidence interval; t = t-value; df = degrees of freedom.

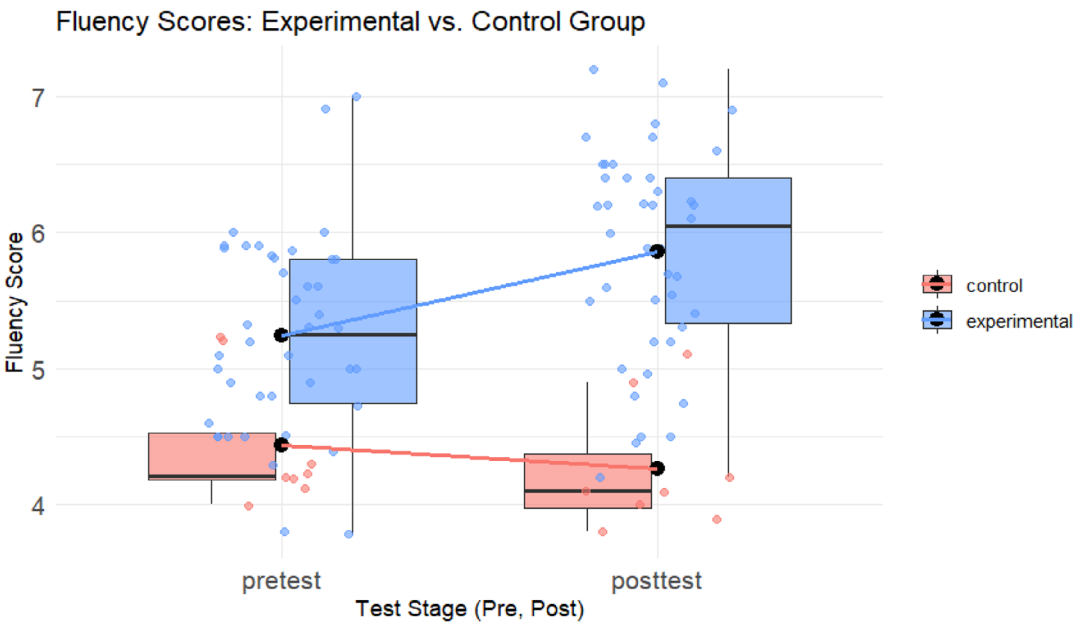

FIGURE 6 | Boxplot of fluency scores in experimental vs. control groups. [Color figure can be viewed at wileyonlinelibrary.com]

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TABLE 8 | Mean comprehensibility scores for Revoicer and Speechify across pre-, post-, and delayed post-tests.

Test stage	Mean (Revoicer)	Mean (Speechify)	Mean difference (*M*diff)	95% CI	*t*-value	df	*p*-value
Pre-test	5.26	5.21	0.05	[−0.33, 0.44]	0.28	33.68	0.780
Post-test	5.81	5.69	0.12	[−0.38, 0.62]	0.49	37.81	0.629
Delayed post-test	5.65	5.66	−0.02	[−0.56, 0.52]	−0.07	23.53	0.948

Note: Mdiff = mean difference; CI = confidence interval; t = t-value; df = degrees of freedom.

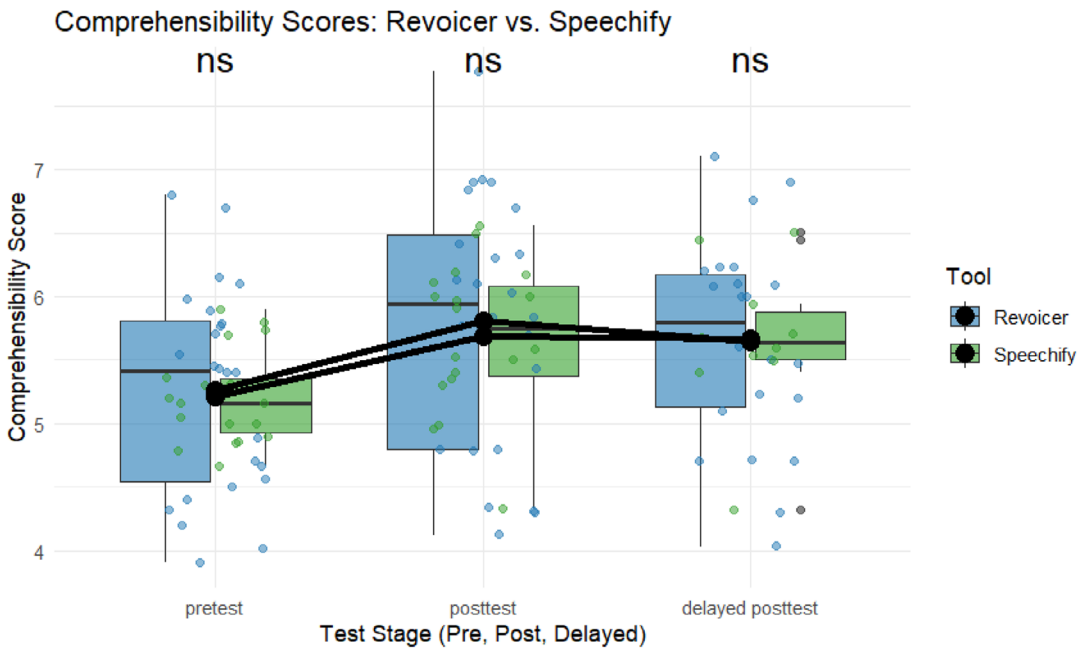

FIGURE 7 | Boxplot of comprehensibility scores for Revoicer vs. Speechify. [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 9 | The results of independent-samples *t*-tests.

Test stage	Mean (Revoicer)	Mean (Speechify)	*t*-value	df	*p*-value	95% CI (lower)	95% CI (upper)
Pre-test	5.27	5.22	0.23	35.65	0.817	−0.37	0.47
Post-test	5.85	5.87	−0.08	38.15	0.934	−0.48	0.44
Delayed post-test	5.74	5.69	0.20	27.67	0.844	−0.41	0.50

Note: t = t-value; df = degrees of freedom; CI = confidence interval.

not possible to state which tool significantly outperformed the other at any time point (see Table 9 and Figure 8). There was no significant difference in L2 fluency in the pre- and post-tests ($p = 0.817$, $p = 0.934$). In the delayed post-test there was no significant retention effect between Revoicer and Speechify ($p = 0.844$).

The findings revealed that regardless of the AI tool used in the phonetic self-imitation practice, this method improves L2 fluency.

5.3 | RQ3: Retention of Improvements Over Time

To answer RQ3, we examined whether the improvements observed after training were maintained in the delayed post-test. As shown earlier in Table 1 and Figure 1, comprehensibility ratings increased significantly from pre-test to post-test and remained higher than in the pre-test in the delayed post-test. Trained sentences continued to be rated higher than untrained ones across all stages (Table 2, Figure 2).

Fluency followed a similar pattern. As presented in Table 4 and Figure 3, fluency ratings improved significantly after training and were partly retained in the delayed post-test. While there was a small decline compared to the immediate post-test, participants still outperformed their pre-test results. Trained sentences were also rated as more fluent than untrained ones in the delayed post-test (Table 5, Figure 4).

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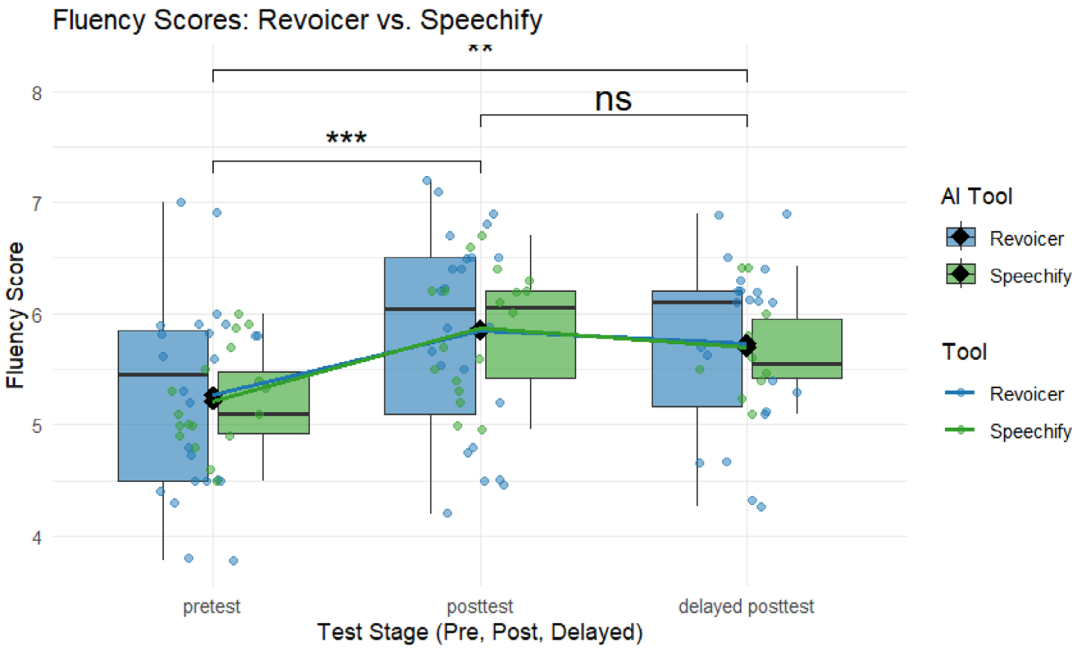

FIGURE 8 | Boxplot of fluency scores for Revoicer vs. Speechify. [Color figure can be viewed at wileyonlinelibrary.com]

6 | Discussion

6.1 | Analysis of the Study

In this study, we looked at the effectiveness of an AI-driven phonetic self-imitation practice on 25 Polish learners' improvement (as measured by ratings of comprehensibility and fluency). Our study also explored whether the effects of pronunciation training sessions would be retained over a longer period, hence delayed post-tests were also applied.

The idea of phonetic self-imitation is an intriguing phenomenon, and understanding why imitating one's voice is more conducive to learning L2 pronunciation than imitating others may lead to further research in this area. One possible explanation for its effectiveness, particularly when learners imitate their AI-cloned voices rather than model voices (e.g., native speakers' voices), lies in the concept of reduced cognitive load. When learners imitate their voice, they may process the input more efficiently due to the familiar tone, timbre, and speech patterns, which allows them to focus more on both segmental and suprasegmental features of L2 pronunciation.

This idea, previously supported by Goldinger and Azuma (2004) and Eger and Reinisch (2019), who showed that episodic memory plays a key role in pronunciation retention, is consistent with our findings. This suggests that L2 learners rely on established articulatory patterns, and familiarity with their acoustic signature may facilitate L2 phonetic perception.

Our results align with Cheung and Babel (2022), who found an own-voice effect in bilinguals. Similarly to the findings of our study, in their research they observed that the participants recognized words faster and more accurately when they were spoken in their own voices. Our findings align with Li and Lian (2022), who demonstrated that imitating a model in one's voice reduces phonological memory load. The improvements in intonation and articulation observed in our study may reflect the cognitive benefits of using a familiar voice. Our results also correspond to Holmes et al. (2018), who argued that one's voice is the most familiar auditory signal. The sustained improvements in comprehensibility in our delayed post-test suggest that familiarity with one's own voice helped retain pronunciation more effectively.

The results of this study showed that both comprehensibility and fluency significantly improved over time. A significant improvement was observed for both trained and untrained sentences/phrases. Although the gains were more prominent for practiced material, the progress seen in untrained sentences indicates that they also significantly improved from pre-test to post-test. This suggests that AI-assisted phonetic self-imitation also supported gains in untrained sentences. Although the ratings for both comprehensibility and fluency in the delayed post-test declined slightly compared to the post-test, they remained higher than in the pre-test. Unlike Derwing and Munro (1997), who noted that gains from pronunciation training diminished quickly, our study indicates that improvements in comprehensibility and fluency were at least partly maintained after four weeks. However, the decline in the delayed post-test in L2 fluency may suggest some loss of fluency gains over time, and that it requires sustained practice to maintain the improvements.

The improvement in L2 comprehensibility and fluency aligns with the findings of Ding et al. (2019), who confirmed that their tool, Golden Speaker Builder, based on the phonetic self-imitation method, contributed to the enhancement of L2 fluency (by focusing on suprasegmental features of pronunciation) as well as comprehensibility. Derwing and Munro (1997) also demonstrated that the comprehensibility of read-aloud speech improved after segmental and prosodic pronunciation training.

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In this study, we used two selected AI voice cloning tools, Revoicer and Speechify, to perform phonetic self-imitation practice. The results showed that there was no significant difference in their effectiveness, suggesting that the phonetic self-imitation method itself can be beneficial, regardless of the two tools used in this study. However, as only two tools were examined, further research is needed to generalize this finding. This immediate implicit corrective feedback, although limited to the model voice, effectively led to improvements in L2 learners' comprehensibility and fluency. The findings align with research on implicit corrective feedback, where exposure to L2 model voices helps L2 learners improve their pronunciation. For instance, Hirose et al. (2003) point out the effectiveness of using learners' voices for providing corrective feedback. This may correspond to AI-based phonetic self-imitation tools employed in the present study. Both Speechify and Revoicer used customized voice cloning to provide individualized feedback reflecting each learner's vocal characteristics, to enable learners to make more targeted improvements in terms of fluency and comprehensibility. This was based on the tone correction system of Peabody and Seneff (2006), where automated adjustments in pitch and intonation facilitated learning prosodic accuracy.

Moreover, the prosody-centered feedback by Pellegrino and Vigliano (2015) has emphasized the significance of personalized training in addressing specific pronunciation challenges. According to their results on Japanese learners of Italian, such customized practice is particularly effective in improving the internalization of native-like prosodic patterns. This applies directly to the findings of this study, since participants who used AI-assisted tools made noticeable improvements in their alignment of pronunciation with native-like models.

Given this, this research indicates that if personalized and adaptive, corrective feedback is effective with traditional systems or through AI-driven tools. It thus supports the conclusion that self-imitation tools, such as Speechify and Revoicer, play a significant role in the continuously evolving corrective feedback system.

6.2 | Limitations

An important limitation in drawing conclusions from this study is that the control group was limited to four people, compared to twenty-one in the experimental group. This may have affected the statistical power of comparisons between the experimental and control groups. Additionally, the comparison between the experimental and control groups should be interpreted with caution, as some differences were already noticeable at the pre-test stage. Another limitation related to this issue is the relatively small number of participants in each subgroup (Revoicer and Speechify). However, although the sample sizes were limited, the implementation of these two tools was intentional. Our goal was to explore whether different AI-based voice cloning tools could influence L2 pronunciation, particularly comprehensibility and fluency. Revoicer was a paid tool, while Speechify offered a free version with optional premium access. Showing that there was no statistically significant difference between a paid and a free tool provides useful information for future research and L2 teachers and learners. However, we are aware that larger sample sizes are needed to confirm these preliminary findings. Although all raters were experienced and provided with detailed written instructions, the absence of a collective calibration session could be considered a limitation.

Another limitation was our focus only on the sentence-level read-aloud task. Although in the pre- and post-tests, participants were asked to answer open-ended questions and their responses were recorded, we decided not to analyze spontaneous speech and instead focused on the read-aloud texts.

Finally, in our study, we did not examine the potential impact of individual preferences or external factors on the process of phonetic self-imitation practice. Participants' feedback on the experiment should also be analyzed.

6.3 | Implications

The research offers practical and educational implications for L2 learners and educators. The AI-driven phonetic self-imitation method allows students to adopt a more flexible approach to improve L2 pronunciation skills at their own pace. Incorporating AI-based phonetic self-imitation tools into curriculum design can complement traditional teaching methodologies with innovative and adaptive strategies.

This integration may offer students more independent and effective learning experiences while allowing teachers to focus on areas requiring in-person support and guidance. These adaptive and personalized capabilities of AI have already been highlighted by Chen et al. (2020), who confirmed the revolutionary impact of L2 pronunciation applications in improving educational outcomes by tailoring learning to students' needs and offering productive, actionable feedback.

7 | Conclusion and Future Directions

The study suggests that AI-driven phonetic self-imitation practice can be helpful for the improvement of L2 pronunciation. It shows how effective phonetic self-imitation training is in enhancing L2 comprehensibility and fluency, regardless of the AI voice cloning tool used. The findings indicate that the improvement was maintained even in the delayed post-test ratings, although a slight decline was observed, particularly in fluency ratings. Trained sentences/phrases were rated higher than untrained ones; however, untrained sentences/phrases also improved, suggesting that AI-driven phonetic self-imitation contributes to overall L2 comprehensibility and fluency improvement.

For future iterations, our plan is to increase the size of the control group. Additionally, while the read-aloud decision ensured controlled and comparable measures across participants, future research should explore more natural speaking contexts to assess whether AI-assisted phonetic self-imitation practice improves spontaneous speech. In the future, we would also like to analyse segmental and suprasegmental improvements. Our goal is to conduct a phonetic analysis to determine whether AI-assisted phonetic self-imitation training primarily improves prosody or also enhances segmental accuracy.

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Acknowledgments

We would like to express our sincere gratitude to the reviewers and editors for their valuable comments and suggestions, as well as to Adrian Smolinski, IT specialist, and to all the students and raters who participated in the study.

Conflicts of Interest

The authors declare no conflicts of interest.

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Appendix 1

Pre-, Post- and Delayed Post-Tests

Sentences

1. The thought of three thousand thirsty thrushes thrusting through thickets.
2. She sells seashells by the seashore, surely she's successful.
3. Peter Piper picked a peck of pickled peppers.
4. How much wood would a woodchuck chuck if a woodchuck could chuck wood?
5. Six slimy snails slid slowly seaward.
6. Red leather, yellow leather, red leather, yellow leather.
7. Betty Botter bought some butter, but she said the butter's bitter.
8. Unique New York, unique New York, you know you need unique New York.
9. I saw Susie sitting in a shoeshine shop.
10. The thirty-three thieves thought that they thrilled the throne throughout Thursday.
11. Susan shines shoes and socks; socks and shoes she shines.
12. Six slippery snails, slid slowly seaward.
13. Rubber baby buggy bumpers bounce.
14. Fuzzy Wuzzy was a bear. Fuzzy Wuzzy had no hair. Fuzzy Wuzzy wasn't very fuzzy, was he?
15. Can you can a can as a canner can can a can?
16. Wayne went to Wales to watch walruses.
17. A black bug bleeds black blood but a blue bug bleeds blue blood.
18. How can a clam cram in a clean cream can?

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19. I wish to wash my Irish wristwatch.
20. Six sharp smart sharks.
21. Truly rural.
22. The great Greek grape growers grow great Greek grapes.
23. She stood on the balcony inexplicably mimicking him hiccupping, and amicably welcoming him in.
24. He threw three free throws.
25. Give Papa a cup of proper coffee in a copper coffee cup.
26. Round the rugged rock, the ragged rascal ran.
27. Quick kiss. Quicker kiss.
28. I scream, you scream, we all scream for ice cream!
29. Eleven benevolent elephants.
30. Bobby Bippy bought a bat.

Text: The sun slowly sets over the serene seaside, casting a golden glow upon the shimmering waves. As the day transitions into night, the gentle breeze whispers through the swaying palm trees, creating a soothing melody that fills the air. The sound of laughter and chatter drifts from nearby cafes, mingling with the rhythmic beat of distant music.

In the distance, a lone seagull soars gracefully across the dusky sky, its wings outstretched against the fading light. With each graceful glide, it embodies the effortless fluidity of flight, a symbol of freedom and tranquillity.

Meanwhile, on the bustling boardwalk, families stroll leisurely, their footsteps echoing against the weathered wooden planks. Children skip and play, their joyful exclamations punctuating the tranquil evening atmosphere.

As the stars begin to twinkle overhead, the soft murmur of the ocean becomes a soothing lullaby, lulling the weary souls into a peaceful slumber. And so, the day surrenders to the night, enveloping the seaside paradise in a blanket of serenity and stillness.

Questions: How has technology changed your life?

Do you think it has made life easier or more complicated?

What technological advancements are you excited about?

Appendix 2

Pronunciation Training Session Material

Week 1

1. The thought of three thousand thirsty thrushes thrusting through thickets.
2. She sells seashells by the seashore, surely she's successful.
3. Peter Piper picked a peck of pickled peppers.

The sun slowly sets over the serene seaside, casting a golden glow upon the shimmering waves. As the day transitions into night, the gentle breeze whispers through the swaying palm trees, creating a soothing melody that fills the air.

Week 2

1. How much wood would a woodchuck chuck if a woodchuck could chuck wood?
2. Six slimy snails slid slowly seaward.
3. Red leather, yellow leather, red leather, yellow leather.

The sound of laughter and chatter drifts from nearby cafes, mingling with the rhythmic beat of distant music. In the distance, a lone seagull soars gracefully across the dusky sky, its wings outstretched against the fading light.

Week 3

1. Betty Botter bought some butter, but she said the butter's bitter.
2. Unique New York, unique New York, you know you need unique New York.
3. I saw Susie sitting in a shoeshine shop.

With each graceful glide, it embodies the effortless fluidity of flight, a symbol of freedom and tranquillity. Meanwhile, on the bustling boardwalk, families stroll leisurely, their footsteps echoing against the weathered wooden planks.

Week 4

1. The thirty-three thieves thought that they thrilled the throne throughout Thursday.
2. Susan shines shoes and socks; socks and shoes she shines.
3. Six slippery snails, slid slowly seaward.

Children skip and play, their joyful exclamations punctuating the tranquil evening atmosphere.

Week 5

1. Rubber baby buggy bumpers bounce.
2. Fuzzy Wuzzy was a bear. Fuzzy Wuzzy had no hair. Fuzzy Wuzzy wasn't very fuzzy, was he?
3. Clever cats chase chubby chipmunks.

The vibrant valley vibrates with vivid vitality, as the verdant vines entwine with the towering trees. As dusk descends, the fiery hues of the horizon highlight the harmonious hues of nature's palette. The babbling brook babbles briskly, its bubbling waters weaving through the winding pathways.

Week 6

1. Fred's friends fried Fritos for Friday's food.
2. Brad's big black backpack broke.
3. The quick brown fox jumps over the lazy dog.

In the twilight, twinkling fireflies flicker, their luminous glow dancing delicately in the dimming daylight. They float with a feather-light finesse, embodying the ethereal essence of the evening.

Week 7

1. Crisp crusts crackle and crunch.
2. Twelve twins twirled twelve twigs.
3. Many mumbling mice are making midnight music in the moonlight.

On the tranquil trail, travelers tread thoughtfully, their whispers weaving a web of warmth and wonder. Children chase cheerful chipmunks, their giggles echoing in the gentle twilight, enhancing the evening's enchanting embrace.

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Week 8

1. Flash messages swiftly shift shapes.
2. Happy hippos hop heartily.
3. Bright blue birds bravely build big nests.

As the moon majestically rises, its silvery sheen casts a serene spell, transforming the tranquil terrain into a dreamy landscape. The distant hoot of an owl adds a haunting harmony, creating a night-time symphony that soothes the soul.

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